

Did COVID Reprice the Commute? Two-Phase Repricing of the Commute Gradient in Nine U.S. Metro Rental Markets

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Abstract

Using a 102,773-row monthly ZIP-code-level rent panel (Zillow Observed Rent Index, 2015–2026) across nine U.S. metropolitan areas, I estimate whether the pre-existing commute gradient repriced across the COVID-19 break in each metro’s covered rental submarket. Per-metro two-way fixed-effects regressions interact three pre-COVID (2019-vintage) gradient measures — average commute time, distance to the central business district, and gravity-model job accessibility — with a two-phase structural break separating the pandemic disruption (2020-03 through 2021-12) from the partial return-to-office era (2022 onward). Where repricing occurred, it favored the periphery: commute-gradient interactions are positive in Phoenix, Denver, Atlanta, and Chicago; job-accessibility interactions are negative in Los Angeles, Dallas–Fort Worth, and Atlanta in both phases, and in the disruption phase in Phoenix and Chicago; Seattle’s repricing runs through CBD distance. With the single exception of Chicago’s job-accessibility channel, no significant disruption-phase effect returns to zero in the return-to-office phase, and in several metros the later-phase point estimates are larger. The paper’s contribution is equal parts finding and discipline: event-study pre-trend checks demote three metros to “trend plus break,” a wild cluster bootstrap at coarse spatial clusters sustains only a handful of metro-channel results, and one metro is reported as uninformative on thin identification. The strictly-clean-and-robust list is short; the directional consistency across nine metros is the stronger result.

1 Introduction

For a century of urban economics, the price of housing has embedded the price of proximity: rents fall with distance from employment centers because commuting is costly, the equilibrium traced by the monocentric bid-rent tradition of Alonso, Muth, and Mills. COVID-19 delivered the sharpest shock that equilibrium has ever received. If work can be done at home, the commute a household avoids by paying central rents is worth less — and the gradient should flatten: peripheral, long-commute locations gain value relative to job-rich cores.

A rapidly assembled literature documented exactly this during the pandemic’s first two years: the “donut effect” of rent and price declines in dense urban cores relative to suburbs (Ramani and Bloom 2021), the flattening of the bid-rent curve across the largest U.S. metros (Gupta, Mittal, Peeters, and Van Nieuwerburgh 2022), and house-price growth concentrated where remote-work exposure was highest (Mondragon and Wieland 2022). The open questions in 2026 are no longer *whether* the disruption happened but *whether it stuck*, *where* it happened, and *how much of the evidence survives honest inference* — questions the return-to-office era makes answerable for the

first time.

This paper asks those questions of nine U.S. metropolitan areas — Atlanta, Chicago, Dallas–Fort Worth, Denver, Los Angeles, Memphis, Miami, Phoenix, and Seattle — using a monthly ZIP-code-level rent panel spanning January 2015 through mid-2026: 62 pre-break and 75 post-break estimation months in every metro. Three design choices distinguish the exercise.

First, **the gradient is measured before the treatment**. The interaction regressors are fixed pre-COVID vintages: average commute time from the ACS 2015–2019 5-year release, geometric distance to the CBD, and log gravity-model job accessibility built from 2019 LODES employment counts. Post-2020 vintages of these same measures are themselves outcomes of the shock (commuting behavior and job geography both moved), and interacting them with a post-COVID dummy would conflate “COVID repriced the pre-existing gradient” with “COVID moved the measured gradient.”

Second, **the break is two-phase**. A single post-COVID dummy averages over a documented non-monotone path — disruption-era repricing followed by partial return-to-office re-steepening — making a near-zero coefficient uninterpretable. Separating Post1 (2020-03 to 2021-12) from Post2 (2022-01 onward) lets the data distinguish persistent repricing from reversal, which is precisely the question the 2026 vantage point makes answerable.

Third, **the honesty rails are co-headline**. Every specification travels with an event-study pre-trend check (was the “repricing” already underway before March 2020?), a wild cluster bootstrap at coarse spatial clusters (are the conventional standard errors understated for spatially smooth regressors?), single-interaction sign checks (is the joint model’s sign collinearity doing the talking?), and explicit identification flags (one metro, Memphis, identifies the break on only 12 ZIP-code areas and is reported as uninformative). The paper’s summary result is deliberately two-sided: the periphery-favoring direction is remarkably consistent across nine metros, and the subset of metro-channel results that survives every check is small.

2 Related work

Ramani and Bloom (2021) coined the donut effect: within large U.S. metros, rents and prices in dense central areas fell relative to suburban rings during 2020–21, with migration flows to match. Gupta et al. (2022) estimated bid-rent functions before and during the pandemic across the 30 largest U.S. metros and found dramatic flattening in both rents and prices, with heterogeneity tied to remote-work prevalence. Mondragon and Wieland (2022) attribute a large share of aggregate 2019–2021 house-price growth to remote work; Howard, Liebersohn, and Ozimek (2023) trace the mechanism through housing demand. Brueckner, Kahn, and Lin (2023) embed work-from-home in a spatial equilibrium model and derive exactly the cross-city and within-city repricing this paper measures. On the measurement side, Dingel and Neiman (2020) classify occupations by teleworkability — the natural exposure measure for a mechanism test this paper defers to future work.

This paper’s marginal contribution to that literature is threefold: (i) a longer window — through mid-2026, spanning four years of return-to-office — where most donut-effect estimates end in 2021–22; (ii) mid-size and Sunbelt metros (Memphis, Phoenix, Denver) alongside the coastal giants that dominate the flattening literature; and (iii) an inference discipline calibrated to the setting: cluster-robust inference at the unit level is known to understate uncertainty for

spatially aggregated regressors (Barrios, Diamond, Imbens, and Kolesár 2012), and few-cluster wild bootstrap methods (Cameron and Miller 2015; Webb 2023) are applied here as a co-equal robustness layer rather than a footnote.

3 Data

3.1 Sources and panel construction

The outcome is the Zillow Observed Rent Index (ZORI) at ZIP-code level: a repeat-weighted index of asking rents across the full rental stock. I use the smoothed **non-seasonally-adjusted** series. The choice is deliberate: Zillow re-estimates seasonal factors over the full sample at each release, so current-vintage pre-2020 seasonally-adjusted values embed post-2020 information — an anticipation artifact located exactly at the break under study. Sample-month fixed effects absorb common seasonality anyway, so seasonal adjustment buys nothing and costs look-ahead. ZIP codes are treated as ZCTAs (ZIP Code Tabulation Areas), a documented approximation inherited from the underlying data pipeline.

The committed panel spans January 2015 through June 2026 — 138 months, 102,773 ZIP-month observations across the nine metros. Because Zillow revises published history between pulls (trailing smoothing re-estimates recent months at each release), the panel is committed behind a *revision gate*: each rebuild replaces the panel wholesale with one coherent Zillow vintage, and the gate quantifies and bounds the revisions between vintages (structural checks fail the build; bounded revisions are reported). Every analysis result is thereby reproducible as of a recorded pull.

The three gradient regressors are fixed pre-COVID measures per ZCTA:

- **Commute time** (`commute_min_proxy_2019`): workers-weighted average one-way commute in minutes, ACS 5-year 2015–2019 release (table B08303), computed at ZCTA geography.
- **CBD distance** (`distance_to_cbd_km`): geometric distance from the ZCTA centroid to the metro’s central business district point (dual-CBD for Dallas–Fort Worth); vintage-free.
- **Job accessibility** (`log_job_accessibility_2019`): a gravity index, $\sum_j \text{jobs}_j \exp(-d_{ij}/10\text{km})$, over the metro’s census tracts, built from 2019 LEHD LODES workplace employment.

An annual LODES accessibility panel (2015–2023) supports the secondary time-varying specifications.

3.2 Coverage and estimand

ZORI publishes a ZIP-month cell only where listing volume clears a threshold. Coverage is therefore selected: 70–96% of each metro’s ZCTAs appear cross-sectionally (Chicago 70%, Atlanta 96%; Table 2), far fewer in the early years (Chicago and Seattle start at 9% of ZCTAs in 2015), and covered areas are systematically larger, denser rental submarkets. Three consequences discipline every claim in this paper. First, all results describe the *covered rental submarket* — no claim extends to uncovered ZCTAs. Second, the unweighted ZCTA-level regressions estimate *average covered-ZCTA* repricing, not renter-weighted repricing (a renter-weighted variant is reported as robustness). Third, “repricing” means the listing index moved — an amalgam of price and listing-composition change; no hedonic adjustment is possible at this altitude. And because

every ZCTA is treated by COVID, there is no control group: estimates are within-metro *relative* descriptions, not causal effects of the pandemic.

Table 2 summarizes the covered samples and the three gradient regressors.

Table 1: **Table 2 — Covered ZCTAs and gradient regressors, mean (SD), computed on each metro’s ZORI-covered ZCTA set.**

Metro	Covered / total ZCTAs	Commute 2019, min	CBD distance, km	Log job access 2019
Phoenix	131 / 150	29.2 (4.5)	29.0 (19.5)	12.00 (1.24)
Los Angeles	246 / 270	34.5 (3.5)	21.8 (12.0)	13.22 (0.60)
Dallas–Fort Worth	177 / 190	30.4 (5.4)	21.6 (14.0)	12.40 (0.80)
Memphis	39 / 52	25.3 (3.9)	20.0 (13.2)	11.58 (0.76)
Denver	90 / 103	29.8 (3.5)	16.7 (10.0)	12.49 (0.61)
Atlanta	112 / 117	34.1 (5.1)	25.3 (14.9)	12.30 (0.74)
Chicago	204 / 291	35.1 (4.5)	32.6 (19.5)	12.76 (0.71)
Seattle	125 / 150	33.7 (4.5)	29.4 (17.9)	12.14 (0.84)
Miami	171 / 180	32.1 (5.5)	44.8 (33.3)	12.31 (0.67)

4 Empirical design

4.1 Specification

All estimation is per metro. The headline specification is a two-way fixed-effects structural break on log rent:

$$\log(\text{zori}_{it}) = \alpha_i + \gamma_t + \sum_x [\beta_x^{P1}(x_i \times \text{Post1}_t) + \beta_x^{P2}(x_i \times \text{Post2}_t)] + \varepsilon_{it}$$

where α_i are ZCTA fixed effects, γ_t are sample-month fixed effects (one dummy per calendar month, absorbing metro-wide shocks and seasonality), x ranges over the three 2019-vintage gradient measures, $\text{Post1}_t = \mathbf{1}[2020-03 \leq t \leq 2021-12]$ (disruption phase) and $\text{Post2}_t = \mathbf{1}[t \geq 2022-01]$ (partial return-to-office phase). Main effects of x_i are absorbed by α_i and phase main effects by γ_t ; only the interactions are identified — which is the question. The final month of each vintage is trimmed (provisional data), leaving 62 pre-break and 75 post-break months in the headline sample; a co-headline variant additionally drops the 2020-03 to 2020-05 transition window (72 post-break months) and leaves every headline coefficient essentially unchanged in all nine metros.

The donut hypothesis predicts $\beta_{\text{commute}}^{P1} > 0$, $\beta_{\text{distance}}^{P1} > 0$, $\beta_{\text{access}}^{P1} < 0$; the Post2 set answers “did it stick?” Because the three gradients are mutually correlated and spatially smooth, each metro also reports three single-interaction models; where joint and single models disagree, the disagreement is collinearity speaking, and the per-metro verdicts lean on whichever pattern is stable across both.

An event study (Spec B) replaces the phase dummies with event-time bins relative to 2020-03 (base bin 2019-03 to 2020-02; 12-month pre bins, 6-month post bins through early 2022, 12-month bins after). Flat pre-break coefficients establish the absence of differential pre-trends along the gradient, so the break coefficients read as departures from a flat pre-path; a drifting pre-path demotes them to “trend plus break,” and the per-metro verdicts report exactly that. (With every ZCTA treated, this check bears on the descriptive reading — trend versus break — not on a causal parallel-trends assumption, which the estimand does not invoke.) Event-study figures carry per-bin identifying-ZCTA counts on a secondary axis, because the earliest bins are thin in several metros (Memphis 7, Denver 12, Seattle 17 ZCTAs).

Two secondary specifications — a time-varying annual accessibility regressor (2015–2023, no carry-forward) and an annual rents-on-lagged-access predictive association with a lead-term falsification — are reported but deliberately kept secondary; the lead-term falsification fails (the *lead* of accessibility is significant) in six of nine metros, so the rents-and-jobs association reads as feedback, not “rents chase jobs.”

4.2 Inference

Estimation is by the within transform with standard errors clustered by ZCTA, robust to the severe serial correlation a smoothed monthly index induces. The small-sample correction deliberately rescales the clustered covariance by $(N - K)/(N - K - G_{\text{absorbed}})$ — the conservative direction relative to the Cameron–Miller/reghdfe convention of omitting absorbed fixed effects from K ; a reviewer should read the slightly inflated standard errors as a choice, not an error.

Unit-level clustering is nonetheless known to understate uncertainty when regressors are spatially smooth (Barrios et al. 2012) — and CBD distance is mechanically so. Each metro therefore also reports wild cluster bootstrap p-values (Cameron and Miller 2015), with Webb weights (Webb 2023) given the small cluster counts, re-clustered at the 3-digit ZIP prefix — spatially coherent USPS sectionals, 4 to 15 per metro. Finally, per-metro identifying counts (n of ZCTAs observed on both sides of the break) are reported with the headline: Memphis identifies the break on only 12 ZCTAs, its conventional cluster t-statistics are oversized, and it is flagged under-identified and read as uninformative throughout.

5 Results

5.1 Two-phase break estimates

Table 1 reports the joint-model interaction coefficients. Units are natural: log-points of rent per commute minute, per kilometer of CBD distance, and per log-point of job accessibility.

Table 2: **Spec A joint model, two-phase break coefficients (2019-vintage regressors).** * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, conventional ZCTA-clustered. “ n ident.” counts ZCTAs observed both pre- and post-break. † Memphis is under-identified (12 ZCTAs); its row is reported for completeness and read as uninformative. Joint Wald tests on each phase’s interaction set reject at $p < 0.0001$ in every metro except Memphis ($p = 0.60 / 0.97$).

Metro	n ident.	Commute P1	Commute P2	Distance P1	Distance P2	Access P1	Access P2
Phoenix	92	0.0037**	0.0048*	−0.0021***	−0.0025**	−0.0259*	−0.0259
Los Angeles	121	0.0016	0.0002	0.0001	0.0006	−0.0582*	−0.0876**
Dallas–Fort Worth	80	0.0010	0.0018	−0.0011***	−0.0011*	−0.0330***	−0.0382***
Memphis †	12	0.0035	0.0008	0.0006	0.0008	0.0575	0.0246
Denver	66	0.0056***	0.0076***	0.0017	0.0026	0.0289	0.0357
Atlanta	86	0.0077***	0.0136***	−0.0032***	−0.0042***	−0.0460***	−0.0711***
Chicago	52	0.0025*	0.0045***	0.0003	0.0023*	−0.0585*	−0.0308
Seattle	71	0.0015	0.0039**	0.0061***	0.0063***	0.0714*	0.0600
Miami	73	−0.0001	−0.0006	0.0012***	0.0008*	0.0084	−0.0079

Three patterns organize the table. **First, where repricing happened, it favored the periphery.** Positive commute interactions (Phoenix, Denver, Atlanta, Chicago), negative access interactions (Phoenix, LA, DFW, Atlanta, Chicago in the disruption phase), and Seattle’s positive distance interactions all point the same way: covered ZCTAs far from jobs gained rent relative to job-rich cores. No metro shows core-favoring repricing that survives both the single-interaction check and the pre-trend check. **Second, the repricing largely did not reverse.** The one clean fade is Chicago’s access effect (−0.0585 to an insignificant −0.0308); elsewhere no significant Post1 effect returns to zero in Post2, and in Atlanta, Denver, Los Angeles (access), and Chicago (commute) the Post2 point estimate exceeds Post1 (point-estimate comparisons; no formal Post2–Post1 test is reported). Four years into return-to-office, the disruption-era reshuffle has stuck or deepened. **Third, the channels differ by metro.** In the joint model, Los Angeles reprices through job accessibility alone (commute and distance conditionally null throughout, in the study’s cleanest event study — though all three of LA’s marginal estimates are periphery-favoring; Table A1); Denver reprices through commute time; Seattle through CBD distance; Miami shows essentially no COVID-specific repricing.

Conditional versus marginal signs. Because the three gradients are mutually correlated, Table 1’s joint-model coefficients are conditional loadings, and the negative distance interactions in Phoenix, DFW, and Atlanta illustrate why that matters: each collapses to approximately zero in its single-interaction model (Post1 marginal estimates +0.0002, −0.0000, and −0.0001, all $p >$

0.59) — conditional artifacts of collinearity, not evidence of core-favoring repricing. The reverse disagreement appears in Seattle, whose access interaction is +0.0714 in the joint model but −0.1080 ($p < 0.0001$) marginally: Table 1’s one apparently significant core-favoring coefficient reverses sign in the single-interaction model. Appendix Table A1 reports the marginal estimates for all nine metros; they are *more* uniformly periphery-favoring than the joint model — job accessibility negative and significant in eight of nine metros (all but under-identified Memphis), commute time positive and significant in seven, distance positive in the five metros where it is significant. The single marginal core-favoring estimate, Miami’s negative commute interaction (−0.0033, $p = 0.0068$), sits on the drifting pre-path that demotes all of Miami’s results.

Magnitudes. Atlanta’s commute $\times$ Post2 coefficient of 0.0136 implies that a covered ZCTA with a 10-minute-longer 2019 average commute gained roughly 14% in relative rent by the return-to-office era ($e^{0.136} - 1$) — with the caveat that Atlanta is a trend-plus-break metro, so part of that accumulation continues a pre-existing steepening. Denver’s 0.0076 implies about 8% per 10 minutes. Scaled by within-metro dispersion (Table 2), a one-standard-deviation commute difference (3.5–5.5 minutes) maps to roughly 2–7% relative rent effects in Post2 across the metros with significant commute repricing — economically meaningful without being implausible.

5.2 Pre-trend verdicts

The event studies discipline the structural-break reading. Los Angeles is the cleanest: all pre-bins statistically indistinguishable from base for all three regressors — the post-break movement is not the continuation of a pre-existing path. Chicago’s commute pre-path sits on zero (a clean break) while its access pre-path drifts. Denver’s commute pre-bins are significantly below base — part of its post-break rise continues a pre-existing steepening. Phoenix (distance) and DFW (distance and access) drift. Atlanta drifts on all three regressors — its large and growing coefficients read as the *acceleration* of a pre-existing periphery-ward steepening, not a clean break. Seattle shows the steepest pre-trends in the study, running straight into its bootstrap-robust distance result. Miami’s tiny distance effect sits on a significant pre-existing path. Memphis’s event study is uninformative (7–12 identifying ZCTAs per pre-bin).

The demotions matter for reading Table 1: Atlanta, Seattle, and Miami’s headline coefficients are “trend plus break,” and Denver’s commute channel, Phoenix’s distance channel, and DFW’s distance and access channels carry partial trend components.

5.3 Spatial-cluster robustness

Re-clustering at the 3-digit ZIP prefix with Webb-weight wild bootstrap thins the significance map considerably. At $p < 0.05$ the bootstrap sustains: Seattle’s *positive* distance interaction in both phases ($p = 0.014 / 0.021$, the study’s most bootstrap-robust result), DFW’s *negative* access interaction in both phases ($p = 0.026 / 0.049$), Denver’s disruption-phase set ($p \leq 0.042$), and the disruption-phase distance interactions in Phoenix ($p = 0.030$) and Atlanta ($p = 0.032$). Two of these demand qualification. The Phoenix and Atlanta distance survivors are the *negative* conditional loadings discussed above — Phoenix’s collapses to null marginally, so it carries no evidential weight on its own, and Atlanta’s marginal distance estimate is likewise null (−0.0001, $p = 0.85$). And Denver’s bootstrap set includes its distance and access channels, which were never conventionally significant (ZCTA-clustered $p = 0.14$ and 0.10) — a case where *coarser* clustering yields *smaller* p-values, the opposite of the understatement concern motivating the

exercise, and a reminder that 4-to-15-cluster bootstrap p-values carry their own finite-sample noise. Everything else — including every Chicago result and LA’s access channel ($p \approx 0.11$ – 0.13) — fails the coarse-cluster bar. Memphis’s ZCTA-level bootstrap is consistent with its uninformative verdict (all $p \geq 0.28$).

The two robustness layers cut against different metros: Seattle is bootstrap-robust but pre-trend-contaminated; Los Angeles is pre-trend-clean but narrowly misses the bootstrap bar; Denver’s commute channel is bootstrap-robust but carries a pre-existing steepening. No metro-channel result clears every check cleanly — which is the honest summary of nine-metro evidence at this spatial resolution — while the directional consistency (periphery-favoring in every metro where anything moves, on both conditional and marginal readings) is the finding that no individual check dislodges.

5.4 Composition and selection

ZCTAs entering the panel after December 2019 are more peripheral than incumbents on all three gradients in seven of nine metros, so incumbent-identified estimates likely *understate* periphery repricing where entrants are peripheral. A balanced-subpanel bound (ZCTAs in-sample by January 2019) sits close to the headline in every metro and changes no substantive conclusion.

6 Robustness

Index choice. Re-running the headline specification for Phoenix, Los Angeles, and Chicago against a local build of the seasonally-adjusted ZORI series moves every interaction coefficient by well under one standard error (the largest shift is 0.18 SE), with no sign changes and no significance crossings. The non-SA choice is not driving the results.

Renter weighting. Weighting ZCTAs by renter prevalence (renter share times population — a different estimand: renter-weighted rather than average-covered-ZCTA repricing) changes no qualitative verdict: strong metros hold or strengthen (Denver commute Post2 0.0076 to 0.0087; Atlanta 0.0136 to 0.0140; Phoenix strengthens on all three regressors), nulls stay null, and no coefficient changes sign. The one notable movement is Chicago’s commute channel, which attenuates in the disruption phase but *strengthens* in Post2 (0.0039, $p = 0.0007$) — Chicago’s return-to-office-era commute repricing is, if anything, concentrated in renter-heavy ZCTAs.

Transition window. Dropping 2020-03 through 2020-05 (the months where trailing smoothing spreads the shock) leaves every headline coefficient essentially unchanged in all nine metros.

Secondary specifications stay secondary. Within-ZCTA annual accessibility is significantly negative in LA, Chicago, and Seattle, positive in Miami, null elsewhere. The rents-on-lagged-access association fails its lead-term falsification in six metros and is reported as feedback, not causation. Mediation shares through contemporaneous job relocation are unstable across metros and are not leaned on.

7 What this paper does not claim

No causal effect of COVID (every ZCTA is treated; no control group). No claim about uncovered ZCTAs — the 4–30% of each metro’s ZIP-code areas, systematically thinner rental markets, that

ZORI does not publish. No decomposition of price versus listing-composition change. No welfare or affordability-burden claims — the outcome is a rent index, not rent-to-income. No cross-metro pooled estimate — per-metro estimation, cross-metro comparison only. And no mechanism test yet: the repricing is *consistent with* the remote-work story, but this paper does not measure teleworkability exposure.

8 Conclusion

Across nine U.S. metros observed through mid-2026, COVID-era repricing of the commute gradient was periphery-favoring wherever it registered, and it largely did not reverse — in several metros the point estimates deepened — through four years of return-to-office. The evidence is strongest as a directional pattern and deliberately modest as a set of individually robust point estimates: pre-trend checks demote three metros to trend-plus-break, coarse-cluster bootstrap inference sustains only a handful of metro-channel results, and one metro is uninformative on thin identification.

The natural next step is a mechanism test: interacting a pre-COVID teleworkability exposure — Dingel–Neiman occupation shares (Dingel and Neiman 2020) applied to each ZCTA’s 2019 LODS industry mix — with the same two-phase break, asking whether the repricing loads on remote-work exposure over and above raw commute geometry. That test, along with Conley spatial-HAC standard errors as a continuous middle ground between the unit-cluster and coarse-cluster inference regimes, and single-family versus multifamily index tiers to separate price from composition, is queued in the project’s public issue tracker.

9 Figures

10 Appendix

Table 3: **Table A1 — Spec A single-interaction (marginal) models: each gradient interacted with Post1/Post2 in its own two-way FE regression.**

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, conventional ZCTA-clustered. † Memphis under-identified. Full per-metro tables with standard errors are in the repository’s `data/processed/<METRO>/rq4_summary_<METRO>.md`.

Metro	Commute P1	Commute P2	Distance P1	Distance P2	Access P1	Access P2
Phoenix	0.0034***	0.0037***	0.0002	−0.0000	−0.0110**	−0.0088
Los Angeles	0.0051***	0.0055**	0.0029***	0.0046***	−0.0636***	−0.0991***
Dallas–Fort Worth	0.0027*	0.0040*	−0.0000	0.0002	−0.0271***	−0.0361**
Memphis	−0.0016	0.0004	−0.0002	0.0003	0.0225	0.0027

†

Metro	Commute P1	Commute P2	Distance P1	Distance P2	Access P1	Access P2
Denver	0.0055***	0.0084***	0.0011**	0.0020**	−0.0192**	−0.0351**
Atlanta	0.0068***	0.0135***	−0.0001	0.0011	−0.0285**	−0.0662***
Chicago	0.0056***	0.0062***	0.0025***	0.0037***	−0.0832***	−0.1189***
Seattle	0.0094***	0.0131***	0.0039***	0.0049***	−0.1080***	−0.1333***
Miami	−0.0033**	−0.0027*	0.0011***	0.0009***	−0.0511***	−0.0481**

11 References

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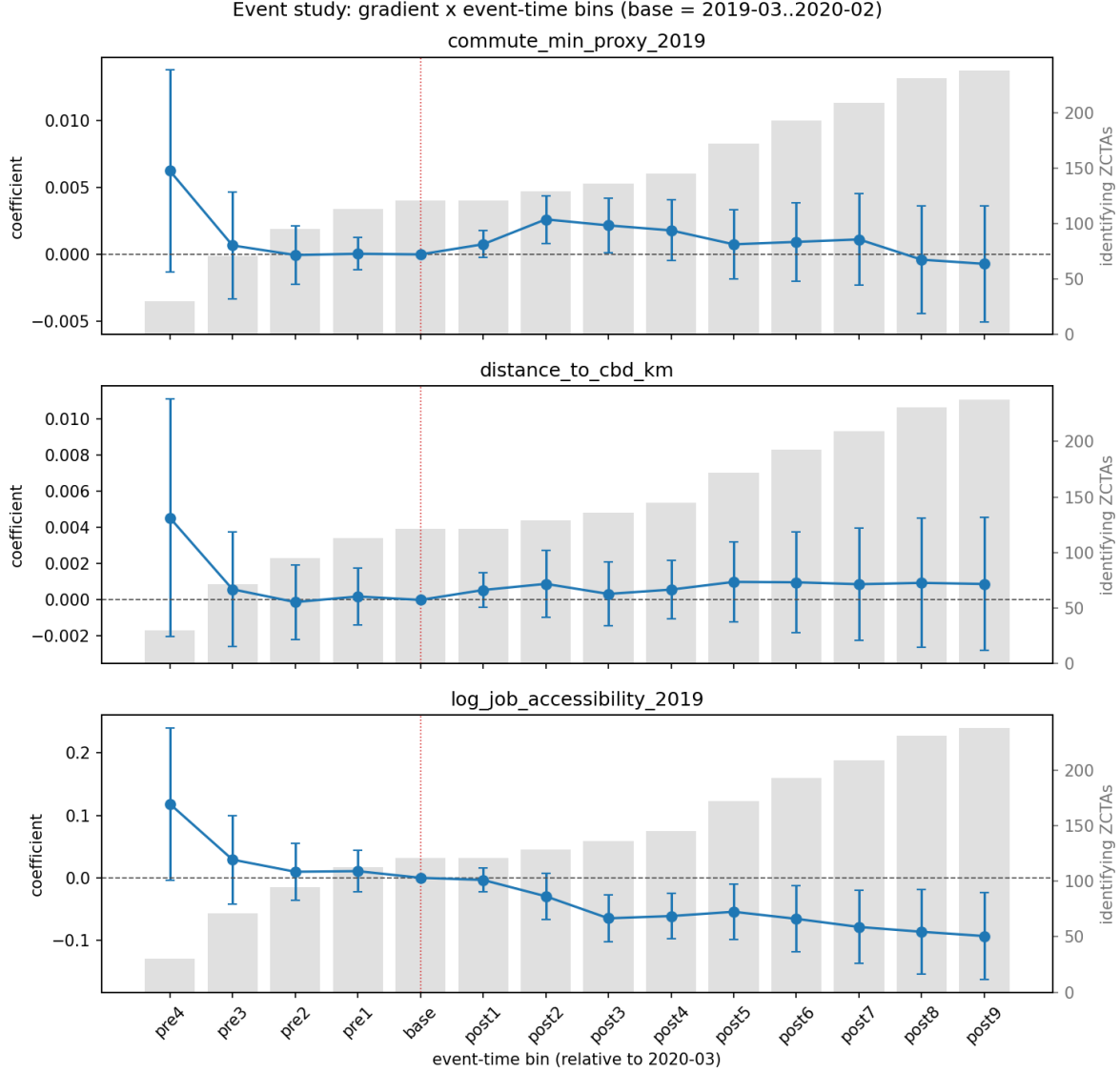


Figure 1: Los Angeles event study — the cleanest pre-trend in the study. Gradient-interaction coefficients by event-time bin (base = 2019-03 to 2020-02); bars show identifying-ZCTA counts per bin. Pre-break coefficients are statistically indistinguishable from base for all three regressors (the elevated earliest bin rests on 30 ZCTAs and stays inside its CI); job-accessibility repricing emerges after 2020-03 and deepens.

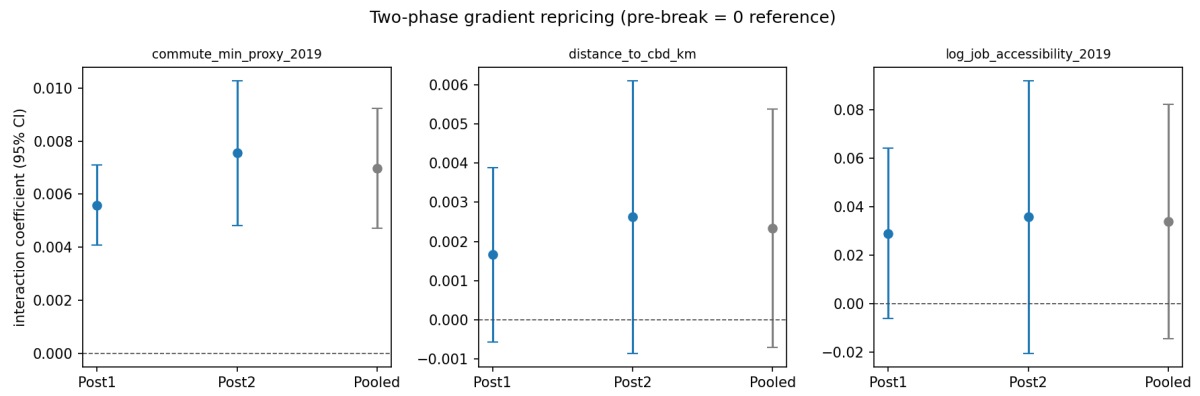


Figure 2: Denver two-phase gradient repricing (pre-break = 0 reference). The commute-gradient interaction is positive in the disruption phase and larger in the return-to-office phase — the study’s strongest commute repricing; distance and access interactions carry wide confidence intervals.